

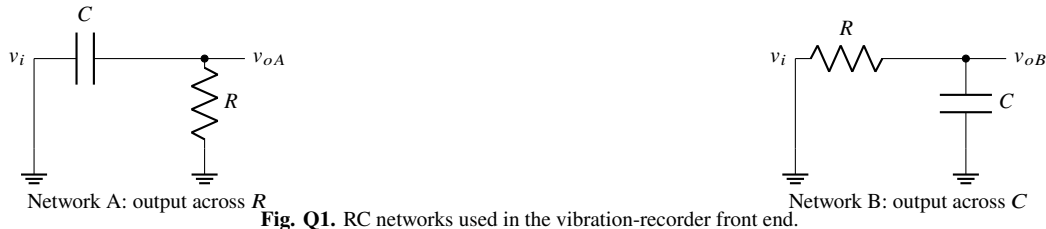
COLLEGE OF ENGINEERING TRIVANDRUM
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
THIRD SEMESTER B. TECH DEGREE
First Internal Exam – August 2026
PCECT303:ANALOG CIRCUITS

Max. Marks: 40

Duration: 2 Hours

Part A – Answer all questions

Scenario for Questions 1–2: RC pulse shaping in a vibration recorder. A bridge-monitoring unit must separate slow structural motion from sharp impact events. The same components, $R = 10\text{ k}\Omega$ and $C = 0.1\text{ }\mu\text{F}$, are connected as shown in Fig. Q1. Network A receives a symmetrical 2 V_{pp} triangular wave at 20 Hz ; Network B receives a zero-mean 1 V_{pp} square wave at 2 kHz .



1. Identify the filter realised by each network and calculate the common cutoff frequency. Using the physical behaviour of a capacitor at very low and very high frequencies, explain which network rejects dc/baseline drift and which smooths rapid variations. State the frequency ranges in which Network A approximates a differentiator and Network B approximates an integrator. [3] [CO1, KL3]
2. For the stated test waveforms, obtain the approximate output waveform, polarity/phase, and peak-to-peak amplitude of each network when it operates in its differentiating or integrating range. Comment briefly on why the approximation is reasonable in each case. [3] [CO1, KL3]

Scenario for Questions 3–4: ADC protection and dc-level restoration. A data-acquisition board accepts signals only between 0.5 V and 4.5 V . One conditioning mode must prevent transient peaks from leaving this range. In another mode, a periodic sensor waveform ranging from -3 V to $+1\text{ V}$ must be shifted into the ADC range (0.5 V – 4.5 V) without changing its shape or peak-to-peak value.

3. Name the nonlinear function required for peak protection. Assuming ideal limiting at 0.5 V and 4.5 V , state the output for inputs of -1 V , 2.5 V , and 6 V . Explain why this operation is not equivalent to ordinary attenuation. [Hint: The requirement is output should not be below 0.5 V or above 4.5 V for any input.] [3] [CO1, KL3]
4. Name the function required for dc-level restoration. Determine the required level shift and the resulting output range. State the clamp polarity, the practical condition on the capacitor time constant relative to the signal period, and how a nonzero diode forward drop would affect the clamped level. [3] [CO1, KL3]

Part B – Answer all questions. Each question carries 14 marks.

5. BJT operating regions, modelling, and Early effect

[14] [CO1, CO2 KL4]

A compact optical receiver uses the NPN transistor Q_1 shown in Fig. Q5 as a voltage-to-current converter. Under normal illumination, the transistor must amplify small detector signals. During a sudden bright flash, however, the increased base voltage may force the transistor out of its intended operating region.

The emitter of Q_1 is grounded, its collector is connected to $V_{CC} = 3.3\text{ V}$ through $R_C = 10\text{ k}\Omega$, and the output v_o is measured at the collector. The base is driven by an ideal voltage source v_B .

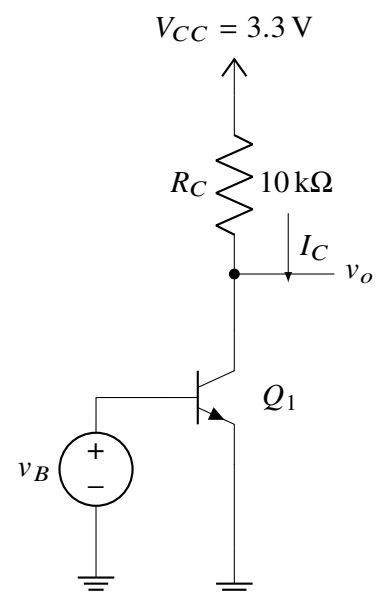


Fig. Q5. Optical-receiver transistor stage.

Use $I_S = 10^{-15}\text{ A}$, $\beta = 100$, and $V_T = 25\text{ mV}$. Unless stated, neglect Early effect. Do not assume the operating region without verifying the junction-bias conditions.

5(a) Large-signal operation and operating-region identification

[7]

- (i) State the base-emitter and base-collector junction-bias conditions for NPN forward-active and saturation operation; also indicate the required ordering of V_C , V_B , V_E . [2]

- (ii) For $V_B = 0.650$ V, calculate I_C , V_C , V_{CE} , and V_{BC} using the forward-active model, and verify the assumed region. [2]
- (iii) A flash raises V_B to 0.720 V. Test the forward-active assumption against the collector load. Identify the actual region and, using $V_{CE,sat} = 0.20$ V, estimate the load-limited I_C and V_{BC} . Explain why $I_C = \beta I_B$ is then invalid. [2]
- (iv) Sketch the I_C - V_{CE} characteristics and load line; mark the normal and flash operating points. [1]

5(b). Small-signal model and Early effect

[7] [CO1,CO2 KL4]

Return to $V_B = 0.650$ V.

- (i) Derive g_m from the forward-active current law; calculate g_m and r_π and state their physical meaning. [2]
- (ii) Draw the low-frequency hybrid- π model without Early effect. For ideal small-signal base drive, find v_o/v_b and explain the phase inversion. [2]
- (iii) With $V_A = 25$ V, determine I_C and V_{CE} self-consistently and verify forward-active operation. [1.5]
- (iv) Calculate r_o , redraw the model including it, and find the revised gain. Explain the physical origin of the Early effect and its influence on the output characteristics. [1.5]

6. Biasing and small-signal analysis of a common-emitter amplifier

[14] CO2, KL4

An S3-ECE student is developing a battery-powered **bridge-vibration monitoring node**. A piezoelectric sensor produces a small sinusoidal voltage with zero dc average. To amplify weak vibration signals, the student connects the sensor through a coupling capacitor directly to the base of an NPN common-emitter stage.

During testing, the oscilloscope shows essentially no amplified output. The student first adds a resistive voltage divider to the base. The transistor then remains strongly driven and the output becomes clipped. A senior engineer finally adds an emitter resistor to obtain the required output.

The successive designs are represented by Fig. Q6. At the signal frequency, coupling capacitors are short circuits. Unless stated, C_E is absent and $V_A = \infty$.

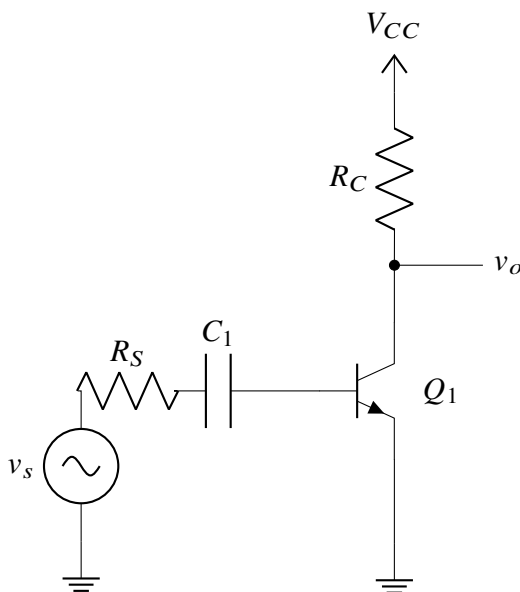


Fig. Q6(a). Initial prototype.

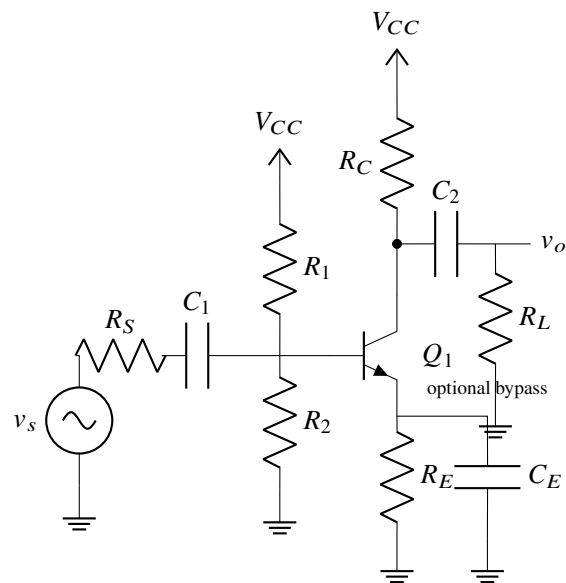


Fig. Q6(b). Divider-biased stage with emitter stabilisation.

Use $V_{CC} = 12$ V, $R_C = 5.6$ k Ω , $R_1 = 68$ k Ω , $R_2 = 15$ k Ω , $R_E = 1$ k Ω , $R_S = 5$ k Ω , $R_L = 10$ k Ω , $\beta = 100$, $V_{BE} = 0.70$ V, and $V_T = 25$ mV. The sensor peak is 10 mV.

6(a). Biasing and operating-point stabilisation

[7] CO2, KL4

- (i) Explain why the initial AC-coupled circuit gives almost no amplified output. How was it solved and how is it helping? [1.5]
- (ii) With the divider present but $R_E = 0$, obtain its Thevenin equivalent, calculate the active-mode operating point, test its physical validity, and estimate the saturation-limited I_C using $V_{CE,sat} = 0.20$ V. Explain why a divider alone need not establish a valid Q-point. [2.5]
- (iii) With $R_E = 1$ k Ω , derive the base-current expression and calculate I_B , I_C , I_E , V_E , V_C , and V_{CE} . Verify forward-active operation. Recalculate I_C for $\beta = 50$ and 200, and explain the stabilising feedback due to R_E . [3]

6(b) Small-signal model, impedances, and gain

[7]

Use the stabilised circuit with $\beta = 100$ and initially no C_E .

- (i) Draw the complete midband hybrid- π model including the divider, R_S , R_E , R_C , and R_L . Calculate g_m and r_π and state their physical meaning. [1.5]
- (ii) Derive the resistance looking into the transistor base; hence find the total input resistance seen by the sensor and the unloaded output resistance. [2]
- (iii) Including source loading and R_L , determine v_o/v_s and the peak output for a 10 mV peak sinusoidal input; state the output polarity. [2]
- (iv) With a large C_E bypassing R_E , recalculate the input resistance and v_o/v_s . Explain the gain-input-resistance trade-off and why the dc Q-point is unchanged. [1.5]

Clearly state all assumptions and verify each transistor operating region.

Relevant Equations

Standard circuit and device relations are collected here for reference. Symbols have their usual meanings.

Circuit relations

$$\begin{aligned}v_R(t) &= Ri_R(t), & i_C(t) &= C \frac{dv_C(t)}{dt} \\v_C(t) &= v_C(t_0) + \frac{1}{C} \int_{t_0}^t i_C(\lambda) d\lambda \\Z_R &= R, & Z_C &= \frac{1}{j\omega C}, & \tau &= RC \\V_o &= V_i \frac{Z_2}{Z_1 + Z_2}, & f_c &= \frac{1}{2\pi RC} \\R_a \parallel R_b &= \frac{R_a R_b}{R_a + R_b}\end{aligned}$$

Ideal and practical diode models

$$\begin{aligned}\text{ON: } v_D &\approx 0 \text{ (ideal),} & v_D &\approx V_\gamma \text{ (constant-drop)} \\ \text{OFF: } i_D &\approx 0\end{aligned}$$

BJT dc relations

$$\begin{aligned}I_C &= I_S e^{V_{BE}/V_T}, & I_B &= \frac{I_C}{\beta}, & I_E &= I_C + I_B \\V_T &= \frac{kT}{q}, & V_{CE} &= V_C - V_E, & V_{BC} &= V_B - V_C \\I_C &\approx I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A} \right)\end{aligned}$$

Small-signal BJT relations

$$\begin{aligned}g_m &= \frac{I_C}{V_T}, & r_\pi &= \frac{\beta}{g_m}, & r_o &\approx \frac{V_A}{I_C} \\i_c &= g_m v_\pi + \frac{v_{ce}}{r_o}\end{aligned}$$

Thevenin equivalent of a divider

$$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}, \quad R_{TH} = R_1 \parallel R_2$$

Basic amplifier and network relations

$$v_o = A_v v_i, \quad A_v = \frac{v_o}{v_i}, \quad R_{in} = \frac{v_i}{i_i}, \quad R_{out} = \frac{v_x}{i_x}$$

$$V = IR, \quad \sum_k i_k = 0, \quad \sum_k v_k = 0$$

Useful logarithmic relations

$$G_{dB} = 20 \log_{10} |A_v|, \quad P_{dB} = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$